

DESIGN OF RUBBLE MOUND BREAKWATERS-2

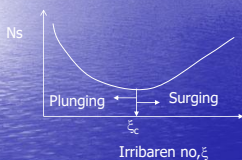
Hudson formula

- Simple and applicable to wide range of armour units

But:

- Regular wave based
- Effect of wave period, storm duration, permeability is not seen in the formula

Van der Meer, 1988



$$\xi = \tan \alpha / \sqrt{s_0}$$

α = slope angle
 s_0 = wave steepness ($= H_0/L_0$)
 H_0 = significant wave height of incident waves at the toe of the structure
 L_0 = deepwater wavelength

Meer's formula (1988)

If $\xi_m < \xi_{mc}$, plunging waves

$$\frac{H_b}{\Delta D_{50}} = 6.2 P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \xi_m^{-0.5}$$

If $\xi_m > \xi_{mc}$, surging waves

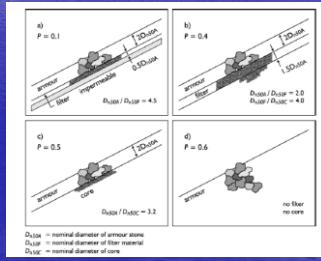
$$\frac{H_b}{\Delta D_{50}} = 1.0 P^{0.13} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \sqrt{\cot \alpha} \xi_m^P$$

$$\xi_{mc} = (6.2 P^{0.31} (\tan \alpha)^{0.5})^{1/(P+0.5)}$$

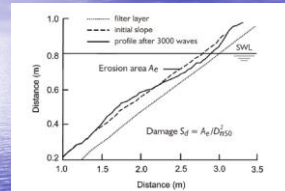
H_b (m)	= Significant wave height at toe of structure	$\xi_m = \frac{\tan \alpha}{\sqrt{s_0}}$
ξ_m	= Surf similarity parameter,	$s_0 = \frac{2H_0}{gT_m^2}$
S_m	= Wave steepness,	
T_m (s)	= Zero up-crossing mean wave period	
α	= Slope angle	
ρ_s (kg/m ³)	= Mass density of stone or unit	
ρ (kg/m ³)	= Mass density of water	
D_{50} (m)	= Normal diameter of stone,	$D_{50} = \left(\frac{W_{50}}{\rho_s} \right)^{1/3}$
W_{50} (kg)	= 50% value of mass distribution curve	
P	= Permeability coefficient of the structure	$S = \frac{A}{D_{50}^2}$
S	= Damage level,	
A (m ²)	= Erosion area in a cross-section	
g (m/s ²)	= Gravitational acceleration	
N	= Number of waves (Storm duration)	

Permeability, P

- Impermeable core, $P=0.1$
- Permeable core+ filter+armour, $P=0.4$
- Permeable core+ armour, $P=0.5$
- Homogenous, $P=0.6$



Damage, S



$$S = A_e / D_{50}^2$$

Initial, (no damage).... $S=2$

Slope (beta)	Damage level		
	Start of damage	Intermediate damage	Failure
1.5	2	3-5	8
2	2	4-6	8
3	2	6-9	12
4	3	8-12	17
6	3	8-12	17

Box 5.13 Design methodology for the Van der Meer formulae

1 Define design wave conditions, H_s and T_m , at the toe of the structure

These may be defined as:

- a single set of wave parameters: H_s and T_m for a chosen extreme return period, eg 100 years
- a set of design wave conditions, each valid for a certain probability of exceedance.

Goda's formula to calculate H_s at any depth

$$H_{1/3} = \begin{cases} K_1 H_0 & d/L_0 \geq 0.2 \\ \min \{ \beta_0 H_0 + \beta_1 h, \beta_{\max} H_0, K_2 H_0 \} & d/L_0 < 0.2 \end{cases}$$

$$\beta_0 = 0.028 (H_0/L_0)^{-0.38} \exp[20(\tan \theta)^{1.5}]$$

$$\beta_1 = 0.52 \exp[4.2 \tan \theta]$$

$$\beta_{\max} = \max \left\{ 0.92, 0.32 (H_0/L_0)^{-0.29} \exp[2.4 \tan \theta] \right\}$$

2 Define acceptable values of damage level parameter, S_d

For extreme conditions it may be acceptable that some damage will occur, whereas only minor damage might be acceptable for less extreme (wave) conditions. This choice should be based on a

3 Determine number of waves, N

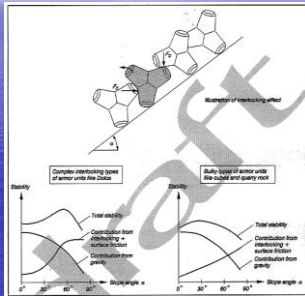
The storm duration gives the number of waves: $N = \text{duration (h)} / T_m \text{ (s)} \times 3600 \text{ (s/h)}$.

4 Determine surf similarity parameter, ξ_m

The surf similarity parameter ξ_m depends on the wave parameters, H_s and T_m , and the slope angle (through $\tan \alpha$). When the choice of the slope angle is free, optimisation of the outcome of the design process is recommended.

Design of Rubble Mound Breakwaters

Slope angle



5 Determine whether waves are *plunging* or *surg*

This is done by calculating the *critical* surf similarity parameter, ξ_{cr} , using Equation 5.138. To solve this equation the structural parameter describing the permeability, P , has to be established (see

6 Determine (average value of the) stability number, $H_o/(\Delta D_{n50})$

7 Determine required armourstone size, D_{n50}

To determine the required armour size, D_{n50} , and hence mass, M_{50} , the mass density of the stone, ρ_s (kg/m^3), is required to calculate the relative buoyant density, Δ . The latter may either be determined or may be prescribed based on a specific rock source for the project.

8 Verification

The outcome of this conceptual design should be verified by performing physical model tests and/or a sufficient safety factor should be taken into account.

example

- Find W of armour stone for the trunk section by Meier formula
- Given: $H_0=2\text{m}$, $T_m=8\text{s}$, $m=1/30$, $d_s=20\text{m}$, $\gamma_s=2.65\text{t/m}^3$, $\gamma_w=1\text{t/m}^3$, $\cot\alpha=2$, $P=0.4$, $N=3000$

If $\xi_m < \xi_{mc}$, plunging waves

$$L_0=100\text{m}, d_s/L_0=0.2,$$

acc to Goda's formula

$$H_s \text{ at str depth} = H_0' K_s$$

$$K_s \text{ (from GWT, } K_s=0.93)$$

$$H_s=2 \times 0.93=1.86\text{m}$$

$$\xi_m = \tan\alpha / \sqrt{s_{0m}} = 0.5 / \sqrt{(1.86/100)} = 3.67$$

$$\xi_{mc} = (6.2 P^{0.31} (\tan\alpha)^{0.5})^{1/(P+0.5)} = (6.2 \times 0.4^{0.31} (0.5)^{0.5})^{1/(0.9)} = 3.77$$

$\xi_m < \xi_{mc}$, plunging waves

example

$$\frac{H_s}{\Delta D_{n50}} = 6.2 P^{0.18} \left(\frac{S}{\sqrt{N}} \right)^{0.2} \xi_m^{-0.5} \quad \frac{1.86}{1.65 D_{n50}} = 6.2 \times 0.4^{0.18} \left(\frac{2}{\sqrt{3000}} \right)^{0.2} 3.67^{-0.5}$$

$$D_{n50}=0.8\text{m}$$

$$W=2.65(0.8)^3=1.33 \text{ ton}$$

Concrete armour units

- damage number N_{ad} : number of displaced armour units within a strip of breakwater slope of width D_n (nominal diameter of armour unit, defined as the equivalent cube size of the unit concerned)
- damage number N_d : number of displaced armour units expressed as a percentage of the total number of armour units placed within a certain range from design water level (a range of $\pm 1.5 H_d$ (design wave height) is typically considered).

$$N_{ad} = \frac{\text{number of units displaced out of armour layer}}{\text{width of tested section} / D_n}$$

Armour type	Damage number	Damage level		
		Start of damage	Intermediate damage	Failure
Cube	N_{ad}	0.2-0.5	1	2
Tetrapod		0.2-0.5	1	1-5
Acropode		0	-	> 0.5
Cube	N_d	-	4%	-
Dolos		0-2%	-	≥ 15%
Acropode		0%	1-5%	≥ 10%

Concrete armour units

Two-layer cubes

For cubes in a double layer on a 1:1.5 slope with $3 < \xi_m < 6$, Equation 5.152, derived by Van der Meer (1988a) based on non-depth-limited wave conditions, gives the relationship between the stability number and the damage number, N_{ad} (-), the wave conditions and the structural parameters.

$$\frac{H_s}{\Delta D_n} = \left(6.7 \frac{N_{ad}^{0.4}}{N^{0.3}} + 1.0 \right) s_{om}^{-0.1} \quad (5.152)$$

where N is the number of waves (-) and s_{om} is the fictitious wave steepness, defined as $2\pi H_s / (g T_m^2)$ (-), based on the mean wave period, T_m (s).



Concrete armour units

Tetrapods

Van der Meer (1988a) presents for tetrapods in a double layer system on a 1:1.5 slope with $3.5 < \xi_m < 6$ and non-depth-limited wave conditions the stability formula, given in Equation 5.153:

$$\frac{H_s}{\Delta D_n} = \left(3.75 \left(\frac{N_{ad}}{\sqrt{N}} \right)^{0.5} + 0.85 \right) s_{om}^{-0.2} \quad (5.153)$$

$$\frac{H_s}{\Delta D_n} = \left(8.6 \left(\frac{N_{ad}}{\sqrt{N}} \right)^{0.5} + 3.94 \right) s_{om}^{0.2} \quad \text{for plunging waves}$$



bottom elevation of cover layer

$$-2H \quad \text{for } d_s > 2H$$

$$\text{to bottom for } d_s \leq 2H$$

toe berm W/10

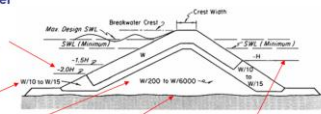
underlayers

$$D_{15, \text{cover}} \leq D_{85, \text{under}}$$

filter layer

$$D_{15, \text{filter}} \leq D_{85, \text{underground}}$$

$$\frac{D_{850a}}{D_{850u}} = 2.2 \text{ to } 2.5$$



$$-H \quad \text{for } d_s > 1.5H$$

$$\text{to bottom for } d_s \leq 1.5H$$

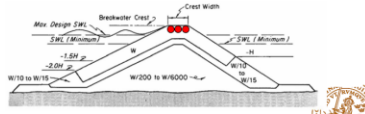
Breakwater Design Elements

* crest width

$$B = nk_{\Delta} \left(\frac{W}{w_r} \right)^{1/3} \quad (n \geq 3)$$

Armor Unit	n	Material	Layer Thickness, w_r	Permeability, k_{Δ}
Quarrystone (smooth) ¹	2	Random	1.00	30
Quarrystone (rough) ²	2	Random	1.00	37
Quarrystone (corrugated) ³	2.5	Random	1.00	44
Other (smooth) ⁴	2	Special	—	27
Quarrystone ⁵	2	Random	1.10	47
Quarrystone ⁶	2	Random	1.08	30
Quarrystone ⁷	2	Random	0.85	49
Quarrystone ⁸	2	Random	1.15	47
Trilobe ⁹	2	Random	1.02	34
Trilobe ¹⁰	2	Random	0.86	30
Trilobe ¹¹	2	Random	1.03	32
Trilobe ¹²	1	Uniform	1.13	47
Quarrystone ¹³	2	Random	—	37

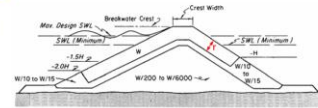
Table 7-13



Breakwater Design Elements

- Layer Thickness

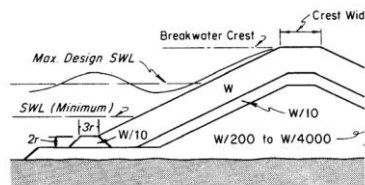
$$r = nk_{\Delta} \left(\frac{W}{w_r} \right)^{1/3}$$



rock units

$$r = \max \begin{cases} 0.3 \text{ m} \\ 2.0 \left(\frac{W_{50}}{w_r} \right)^{1/3} \end{cases} \quad (7-123)$$

$$1.25 \left(\frac{W_{\max}}{w_r} \right)^{1/3} \quad (7-124)$$



$$3r = 3k_{\Delta} \left(\frac{W_{50}}{w_r} \right)^{1/3} \quad \text{where } W_{50} = W/10$$

Toe and scour protection

In most cases the armour of the front face of a rubble mound breakwater and other rock-armoured structures is protected near the bottom by a toe, see Figure 5.72.

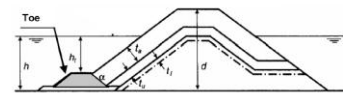


Figure 5.72 Typical cross-section with toe protection

Table 5.46 Stability of toe protection

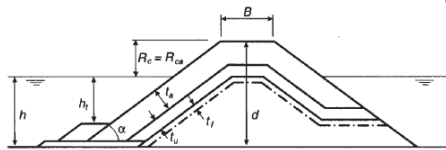
h_t/h	H_d/AD_{100}
0.5	3.3
0.6	4.5
0.7	5.4
0.8	6.5

Breakwater Design Elements

Elevation of the core +1, +1.5m

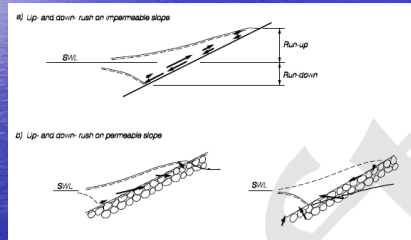
Elevation of the filter layer= core+2Dnfilter

Elevation of the crest (determined by run-up height)



Wave Run-up

- Wind-generated waves have wave periods which trigger wave breaking on almost all sloping structures.
- The wave breaking causes runup, Ru , and rundown, Rd , defined as the maximum and minimum water-surface elevation measured vertically from the still-water level (SWL),

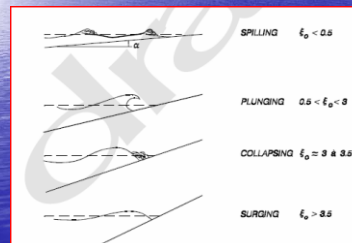


Wave Run-up

- Ru and Rd depend on the height and steepness of the incident wave, as well as the slope angle, the surface roughness, and the permeability and porosity of the slope.
- Maximum values of flow velocities and values of Ru and Rd for a given sea state and slope angle are reached on smooth impermeable slopes.

Wave Run-up

Wave runup and rundown on a structure depend on the type of wave breaking. Breaker types can be identified by the so-called surf-similarity parameter, ξ (Battjes 1974b). The parameter ξ is also referred to as the breaker parameter or Iribarren number.



$$Ru = \frac{a\xi}{1+b\xi} H$$

$$\xi = \frac{\tan \alpha}{\sqrt{H/L_0}}$$

$$a=0.8$$

$$b=0.5$$

(c) For irregular waves the surf-similarity parameter is defined as

$$\xi_{\text{dem}} = \frac{\tan \alpha}{\sqrt{s_{\text{dem}}}} \quad \text{or} \quad \xi_{\text{top}} = \frac{\tan \alpha}{\sqrt{s_{\text{top}}}}$$

where

$$s_{\text{dem}} = \frac{H_s}{L_{\text{dem}}} = \frac{2\pi H_s}{g T_m^2}$$

$$s_{\text{top}} = \frac{H_s}{L_{\text{top}}} = \frac{2\pi H_s}{g T_p^2}$$

and

H_s = significant wave height of incident waves at the toe of the structure

T_m = mean wave period

T_p = wave period corresponding to the peak of the wave spectrum

Wave Run-up

Geçirilmiş Eğimli Taş Kaplamalı Yüzeylerde Dalga Tırmanması (CEM, 2003)

$$\left. \begin{aligned} R_{wi} \% / H_s &= A \xi_{\text{dem}} & 1.0 < \xi_{\text{dem}} \leq 1.5 \\ R_{wi} \% / H_s &= B (\xi_{\text{dem}})^C & 1.5 < \xi_{\text{dem}} \leq (D/B)^{1/C} \\ R_{wi} \% / H_s &= D & (D/B)^{1/C} \leq \xi_{\text{dem}} < 7.5 \end{aligned} \right\} \quad (2.7)$$

Denklemler (2.7) de kullanılan katsayılar Tablo 2.3'den alınır.

Tablo 2.3 Geçirilmiş ve Geçirimsiz Taş Kaplamalı Eğimlerde Kullanılacak Katsayılar (CEM, 2003)

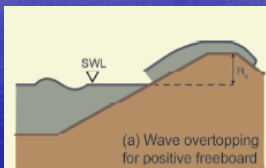
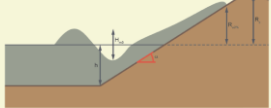
%H	A	B	C	D ¹
0.1	1.12	1.34	0.55	2.58
2.0	0.96	1.17	0.46	1.97
5	0.86	1.05	0.44	1.68
10	0.77	0.94	0.42	1.45
33 (belirgin)	0.72	0.86	0.41	1.35
50 (ortalama)	0.47	0.60	0.34	0.82

$R_{wi} \%$ = runup level exceeded by i percent of the incident waves

Wave Run-up-----overtopping

if $R_u > R_c$

R_{run} = Wave runup height
 R_{top} = Freeboard
 H_{top} = Wave height at the toe of the structure
 L_{top} = Wave length at the toe of the structure



Wave overtopping



17 Severe wave overtopping at Salthouse Hoe, UK

Wave overtopping



Wave overtopping



23 Wave overtopping onto a moving train, UK

Wave overtopping



Green water overtopping Howth, UK



22 Spray overtopping at the breakwater at Margate, UK

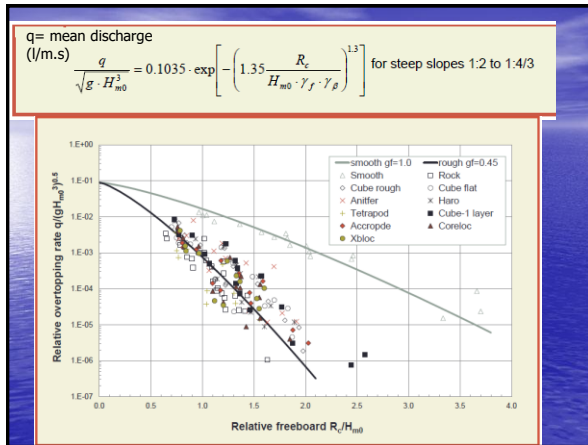
Wave overtopping

EurOtop

Manual on wave overtopping of sea defences and related structures

An overtopping manual largely based on European research, but for worldwide application

Second Edition



$$\frac{q}{\sqrt{g \cdot H_{m0}^3}} = 0.1035 \cdot \exp \left[- \left(1.35 \frac{R_c}{H_{m0} \cdot \gamma_f \cdot \gamma_\beta} \right)^{1.3} \right] \text{ for steep slopes 1:2 to 1:4/3}$$

Type of armour layer	γ_β
Smooth impermeable surface	1.00
Rocks (1 layer, impermeable core)	0.60
Rocks (1 layer, permeable core)	0.45
Rocks (2 layers, impermeable core)	0.55
Rocks (2 layers, permeable core)	0.40
Cubes (1 layer, flat positioning)	0.49
Cubes (2 layers, random positioning)	0.47
Antifers	0.50
HARO's	0.47
Tetrapods	0.38
Dolosse	0.43
Accropode™ I	0.46
Xbloc®, CORE-LOC®, Accropode™ II	0.44
Cubipods one layer	0.49
Cubipods two layers	0.47

$$\frac{q}{\sqrt{g \cdot H_{m0}^3}} = 0.1035 \cdot \exp \left[- \left(1.35 \frac{R_c}{H_{m0} \cdot \gamma_f \cdot \gamma_\beta} \right)^{1.3} \right] \text{ for steep slopes 1:2 to 1:4/3}$$

γ_β = reduction for the wave incidence angle
if the wave attack is perpendicular, ($\gamma_\beta = 1$),

$$\gamma_\beta = 1 - 0.0063|\beta| \text{ for } 0^\circ \leq |\beta| \leq 80^\circ$$

for $|\beta| > 80^\circ$ the result $\beta = 80^\circ$ can be applied

Neural network tool to predict overtopping amount

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Overtopping Neural Network

To all software

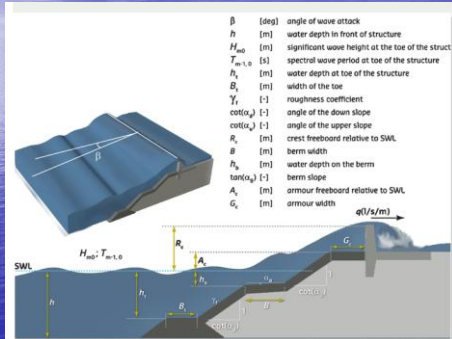
General

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For the design, safety assessment and rehabilitation of coastal structures reliable predictions of wave overtopping are required. Overtopping Neural Network is a

Neural network tool to predict overtopping amount



Tolerable wave overtopping

Table 3.1 Limits for wave overtopping for structural design of breakwaters, seawalls, dikes and dams

Hazard type and reason	Mean discharge q (l/s per m)	Max volume V_{max} (l per m)
Rubble mound breakwaters; $H_{w0} > 5$ m; no damage	1	2,000-3,000
Rubble mound breakwaters; $H_{w0} > 5$ m; rear side designed for wave overtopping	5-10	10,000-20,000
Grass covered crest and landward slope; maintained and closed grass cover; $H_{w0} = 1 - 3$ m	5	2,000-3,000
Grass covered crest and landward slope; not maintained grass cover, open spots, moss, bare patches; $H_{w0} = 0.5 - 3$ m	0.1	500
Grass covered crest and landward slope; $H_{w0} < 1$ m	5-10	500
Grass covered crest and landward slope; $H_{w0} < 0.3$ m	No limit	No limit

Tolerable wave overtopping

Table 3.3 Limits for overtopping for people and vehicles

Hazard type and reason	Mean discharge q (l/s per m)	Max volume V_{max} (l per m)
People at structures with possible violent overtopping, mostly vertical structures	No access for any predicted overtopping	No access for any predicted overtopping
People at seawall / dike crest. Clear view of the sea.		
$H_{w0} = 3$ m	0.3	600
$H_{w0} = 2$ m	1	600
$H_{w0} = 1$ m	10-20	600
$H_{w0} < 0.5$ m	No limit	No limit
Cars on seawall / dike crest, or railway close behind crest		
$H_{w0} = 3$ m	<5	2000
$H_{w0} = 2$ m	10-20	2000
$H_{w0} = 1$ m	<75	2000
Highways and roads, fast traffic	Close before debris in spray becomes dangerous	Close before debris in spray becomes dangerous

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Effect of the armor crest freeboard relative to the crown wall freeboard on wave overtopping for simple rubble mound slopes

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ABSTRACT

Several studies have been carried out to investigate the effect of crest parameters on the wave overtopping for armored slopes with crown walls. However, the effect of the armor crest freeboard is still under question. In this study, for the first time, a series of hydraulic model experiments are conducted specifically to investigate how the armor crest freeboard relative to the crown wall freeboard affects the wave overtopping rate. Experimental results indicate that while the armor crest freeboard lower than the crown wall freeboard is giving larger overtopping, higher armor crest freeboard is reducing the overtopping. However, this reduction is not same as the reduction due to the increase in the crown wall freeboard. The reason may be the presence of the armor crest. For the first time, a new correction factor is proposed to describe the change in the wave overtopping due to the armor crest freeboard by using experimental results. The correction factor C_{ac} is applied to cover the effect of armor crest freeboard in the predictions of Eurtop (2018). The verification study presents that overtopping rate predictions of corrected Eurtop (2018) are more consistent with the measured rate results compared to the predictions of the original formula, if the armor crest freeboard is not equal to the crown wall freeboard.

Crown walls

A superstructure consisting of a concrete cap block or wave wall is named the crown wall.

The crest level to be considered for overtopping is the crest of the crown wall.

width is determined with the same considerations as for rubble mounds, although the required dimensions of the crown wall to prevent sliding or overtopping may dictate the wall

- a crown wall with parapet may lead to a substantial reduction in the amount of armourstone that would otherwise be needed for a comparable conventional design (see Figure 6.23)
- if overtopping is allowed, the crown wall may limit the width of the mound and by its shape protect the lee-side slope (see Figure 6.23)
- access may be required along the breakwater for port operations or for recreation.

Where berths are constructed immediately behind a breakwater, it is common practice for the crown wall to carry facilities for cargo loading and unloading (pipelines, conveyor systems) and electrical and water supply systems.

Crown walls

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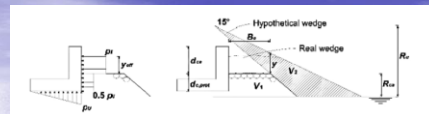
width is determined with the same considerations as for rubble mounds, although the required dimensions of the crown wall to prevent sliding or overtopping may dictate the wall



There are certain disadvantages related to a crown wall, which should be taken into account in selection and design:

- the crown wall represents a rigid element in a structure that is flexible by nature; uneven settlements may lead to structural problems for the elements of the crown wall and for any facilities located on the superstructure
- reduced interlocking of the upper row of armour that is placed immediately adjacent to the crown wall
- the tendency to increase the parapet wall in order to reduce the volume of armourstone can lead to very large wave impact forces on this wall
- the reduction of overtopping by a crown wall increases wave attack on the armour layer
- the crown wall increases the risk of excessive pore pressures in the mound
- overtopping water becomes concentrated into a jet, and can be a potential danger for the lee-side armour if the slab is not wide enough. Chute blocks at the inner (harbour side) edge of the slab can break the jet (see special note in the introduction of Section 5.2.2.12 and Figure 6.25c).
- higher cost and construction time compared with conventional rubble mound
- disruption in the armour layer construction sequence and increased risk of damage until the primary armour layer is completed.

Design of crown wall (Pedersen, 1996)

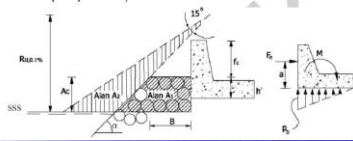


Pedersen determined that the maximum wave loads are caused by wave impact. First the impact pressure on the unprotected crown wall is calculated. Secondly, this value is reduced to obtain the pressure for the protected part of the crown wall. Both pressures are integrated to calculate the horizontal force.

Pedersen assumes that the horizontal pressure on the vertical wall protected by the armour units is 50% of the maximum impact pressure.

Coefficient	Definition	Unit
$P_i = R_i P_{u,i} (R_{i,0.1\%} - R_{i,5\%})$		
y_{ef}	Effective height of impact zone	m
y	Thickness of hypothetical run up wedge	m
d_{un}	Height of unprotected part of the crown wall	m
$y = \begin{cases} \frac{R_{u,0.1\%} - A_e}{\sin \alpha} \frac{\sin 15^\circ}{\cos(\alpha - 15^\circ)} & y > 0 \\ 0 & y \leq 0 \end{cases}$		

Design of crown wall (Pedersen, 1996)

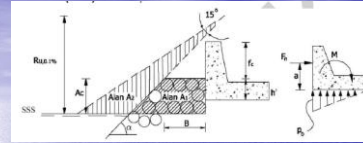


$$F_{h,0.1\%} = 0.21 \sqrt{\frac{L_{om}}{B}} \left(1.6 p_m y_{eff} + A \frac{p_m}{2} h' \right) \quad (VI-5-187)$$

$$M_{0.1\%} = a \times F_{h,0.1\%} = 0.55 (h' + y_{eff}) F_{h,0.1\%} \quad (VI-5-188)$$

$$p_{h,0.1\%} = 1.00 A p_m \quad (VI-5-189)$$

where: $F_{h,0.1\%}$ Horizontal wave force per running meter of the wall corresponding to 0.1% exceedance probability
 $M_{0.1\%}$ Wave generated turning moment per running meter of the wall corresponding to 0.1% exceedance probability
 $p_{h,0.1\%}$ Wave uplift pressure corresponding to 0.1% exceedance probability
 L_{om} Deepwater wavelength corresponding to mean wave period
 B Berm width of armor layer in front of the wall
 p_m $p_m = \rho_w g (H_{h,0.1\%} - A_c)$
 $H_{h,0.1\%}$ Wave runup corresponding to 0.1% exceedance probability



α Slope angle of armor layer

A_c Vertical distance between MWL and the crest of the armor berm

$A = \min\{A_2/A_1, 1\}$, where A_1 and A_2 are areas shown in the figure

$y_{eff} = \min\{y/2, f_c\}$

$$y = \begin{cases} \frac{R_{h,0.1\%} - A_c}{20 \sin \alpha} \frac{\cos 15^\circ}{\cos(\alpha - 15^\circ)} & y > 0 \\ 0 & y \leq 0 \end{cases}$$

h' Height of the wall protected by the armor layer

f_c Height of the wall not protected by the armor layer

$$f(F_G - F_U) \geq F_H \quad \text{for stability against sliding} \quad (5.199)$$

where:

F_G = (buoyancy-reduced) weight of the crown wall element (N), $= (M_{cw} - V_{cw} \rho_w) g$, where M_{cw} and V_{cw} are the mass and the volume of the crown wall

F_U = wave-induced uplift force (N)

F_H = wave-induced horizontal force (N)

f = friction coefficient (-).

$$FS_{SLIDING} = \frac{\mu(W - U)}{p}$$

$$FS_{OVERTURNING} = \frac{Wt - M_U}{M_p}$$

$$M_G - M_U \geq M_H \quad \text{for stability against overturning}$$

where:

M_G = stabilising moment due to mass of the crown wall element (Nm)

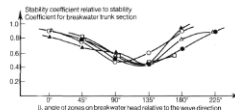
M_U = wave-generated moment due to uplift force (Nm)

M_H = wave-generated moment due to horizontal force (Nm).

Breakwater roundheads

Breakwater roundheads involve a special physical process, as wave breaking over roundheads yields large velocities and wave forces. For a specific wave direction only a limited area of the roundhead is exposed to high wave attack. This area around the still water level, about 120°–150° from the wave direction and thus on the lee side of the roundhead, is shown in Figure 5.88. To obtain the same stability as for the trunk section two options are available (which may be combined):

- to increase the mass of the armourstone (by larger units and/or higher mass density)
- to make the side slope of the roundhead less steep.



Legend:

Wave directions:

50°

90°

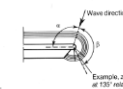
130°

150°

Note:

Slope 1:1.33

Wave steepness $H_w = 0.025$



Example zone at 135° relative to wave direction

Table 5.52a Hudson stability coefficients, K_D , for no damage and minor overtopping

Material (+ slope)		Trunk		Roundhead	
		Breaking wave	Non-breaking wave	Breaking wave	Non-breaking wave
Armourstone, randomly placed	(1:1.5)	2.0	4.0	1.9	3.2
	(1:2.0)	2.0	4.0	1.6	2.8
	(1:3.0)	2.0	4.0	1.3	2.3
Tetrapods	(1:1.5)	7.0	8.0	5.0	6.0
	(1:2.0)	7.0	8.0	4.5	5.5